

Timothy works at the regional Integrated System Operator (ISO). The ISO operates a nodal transmission market. He is responsible for ensuring that all market participants have equal and fair access to the market. This requires two main functions: 1) evaluating proposals from transmission developers to address known issues, and 2) processing the generation interconnection requests in the ISO footprint.

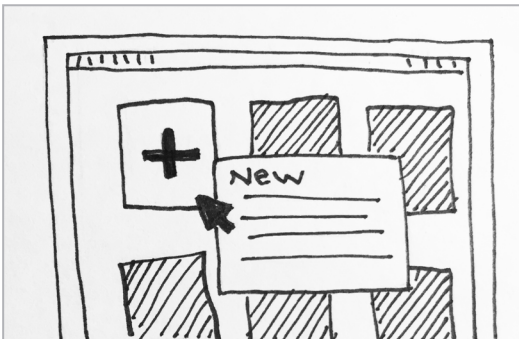


New project

1

Timothy needs to study the transmission system within the ISO footprint as well as interconnections to neighboring systems.

+ *Receives a new assignment*



Create study

2

He creates a new study and uses **the nodal simulation ready data from Energy Exemplar** with the **current planning load flow for future years** as a starting point.

- + *Clicks 'New Study'*
- + *Wizzard overlay*
- + *Selects a study template*
- + *Imports/opens latest data set*
- + *Imports planned load flow*
- + *Imports Tim's email attachment*



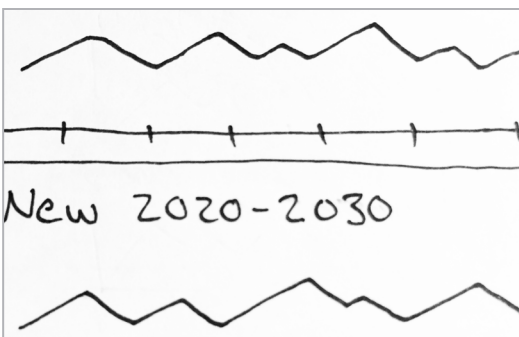
Collaboration

3

Timothy **collaborates with his colleagues** and **updates the ISO vetted data as needed.**

Data will include: accurate operating constraints, demand profiles, fuel price forecasts, renewable profiles, etc.

- + *Shares data with his colleagues*
- + *Updates properties as needed*



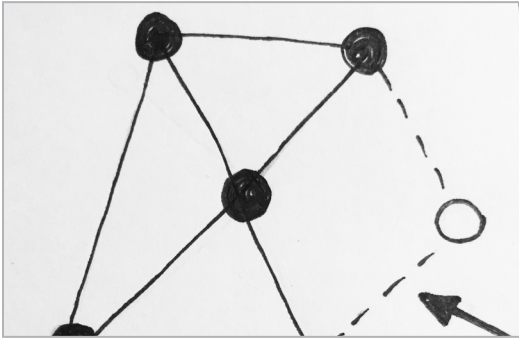
Run scenarios

4

Timothy starts off by **evaluating proposals from transmission developer** to address known issues.

Proposals can come from member utilities or transmission developers – Timothy needs to be agnostic to the source of the proposals.

- + *Starts reviewing proposals*
- + *Runs scenarios of all proposals*



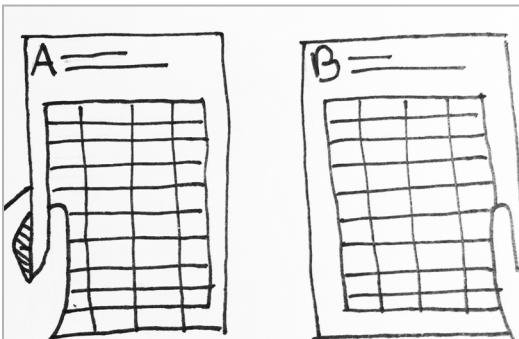
Identify congestions

5

Timothy investigates how well the proposal addresses known **congestion issues**.

Each proposal is run against the future system independently to understand how well it addresses known issues.

- + Compares results w/ future system
- + Opens geospatial view
- + No congestion issues are indicated



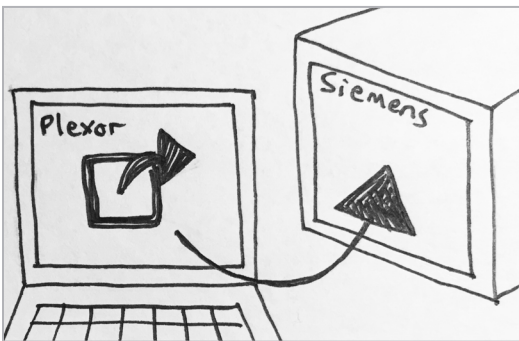
Iterate time frames

6

Timothy runs **multiple future years different scenarios on the base data**.

The metrics used are a combination of economics and reliability – the economics will include the cost of the projects and the benefit in production cost realized by the system, while the reliability metrics will include expected overloads on the system and expected voltage conditions.

- + Runs scenarios with different time frames



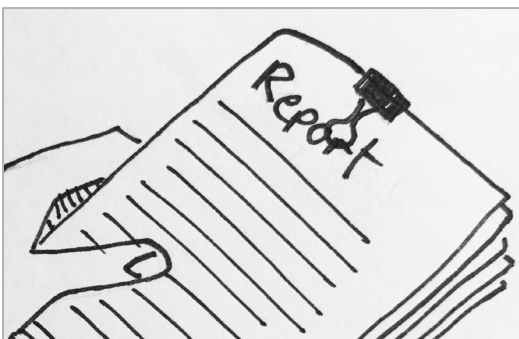
Export and stress test

7

Timothy tests a limited set of scenarios that reflect 'stress' cases for the system in a power flow software such as PSS-E or PowerWorld to **ensure the economic study meets reliability standards**.

These cases usually consist of peak demand hours as well as some high or low generation levels from key facilities. This process is done externally - the integration with Siemens PSS-E will improve this process dramatically.

- + Exports scenario data
- + Opens 3rd party tool
- + Imports and validates stress cases



Submit report

8

Timothy ranks proposals by their metrics and **submits them to the ISO committee** for their approval process.

Timothy then analyzes new generation requests in the same manner as above, and ranks/submits them as well.

- + Creates the report
- + Submits report to ISO committee

Timothy's Narrative

ISO Narrative - Timothy

Timothy works at the regional Integrated System Operator (ISO). The ISO operates a nodal transmission market. He is responsible for ensuring that all market participants have equal and fair access to the market. This requires two main functions: 1) evaluating proposals from transmission developers to address known issues, and 2) processing the generation interconnection requests in the ISO footprint.

Timothy has to study the transmission system within the ISO footprint as well as interconnections to neighboring systems.

He will **use the nodal simulation ready data from Energy Exemplar** with the **current planning load flow for future years** as a starting point, and will **update some of this with ISO vetted data via internal collaboration**. This data will include but not be limited to: accurate operating constraints, demand profiles, fuel price forecasts, renewable profiles, etc.

Timothy will **start off by evaluating proposals from transmission developers** to address known issues. These proposals can come from member utilities or transmission developers – Timothy needs to be agnostic to the source of the proposals.

Each proposal will be run against the future system independently to understand how well it addresses known issues. His objective is to capture how well the proposal addresses known congestion issues.

The metrics used are a combination of economics and reliability – the economics will include the cost of the projects and the benefit in production cost realized by the system, while the reliability metrics will include expected overloads on the system and expected voltage conditions.

Timothy will **run multiple future years different scenarios on the base data**

Timothy will also need to test a limited set of scenarios that reflect 'stress' cases for the system in a power flow software such as PSS-E or PowerWorld to **ensure the economic study meets reliability standards**. These cases usually consist of peak demand hours as well as some high or low generation levels from key facilities. This process is done externally - the integration with Siemens PSS-E will improve this process dramatically.

Timothy will rank these proposals by their metrics and **submit them to the ISO committee** for their approval process. Timothy then analyzes new generation requests in the same manner as above, and ranks/submits them as well.

Below is a statement of responsibility for the SPP Transmission Committee:

Southwest Power Pool Transmission Committee:

The Economic Studies Working Group (ESWG) advises and assists Southwest Power Pool (SPP) Staff in the determination of the appropriate data, planning methods and processes to be used in the development and evaluation of economic transmission expansion options for the SPP region. The ESWG will be responsible for data, sources, models, economic planning methodology, timing and implementation, application, economic parameters and metrics to be used in the development and evaluation of economic expansion options in the SPP region. The ESWG will also review the economic planning processes used by SPP Staff and offer proposals for improvement of these processes. The ESWG will be responsible for all economic modeling parameters and data for the SPP region, and will conduct as needed updates to all data for the stakeholders on a regular basis as appropriate for regional planning needs. Additionally, the ESWG will have oversight on appropriate access of data to third party consultants.